

EXERCISE-3

Part 1: Comparison and Logical Operators

Q1. Retrieve all products with a unit price greater than 20.

QueryQuery History

123

SELECT *
FROM products
WHERE unit_price>20

Data OutputMessagesNotifications

Q2. Fetch all employees whose birth date is before '1960-01-01'.

Query

Query History

1

2

3

SELECT *

FROM employees

WHERE birth_date < '1960-01-01'

Data Output

Messages

Notifications

SQL

	employee_id [PK] smallint	last_name character varying (20)	first_name character varying (10)	title character varying (30)	title_of_courtesy character varying (25)	birth_date date	hire_date date
1	1	Davolio	Nancy	Sales Representative	Ms.	1948-12-08	1992-05-01
2	2	Fuller	Andrew	Vice President, Sales	Dr.	1952-02-19	1992-08-14
3	4	Peacock	Margaret	Sales Representative	Mrs.	1937-09-19	1993-05-03
4	5	Buchanan	Steven	Sales Manager	Mr.	1955-03-04	1993-10-17
5	8	Callahan	Laura	Inside Sales Coordina...	Ms.	1958-01-09	1994-03-05

Q3. Get a list of customers who are from either Germany, France, or the UK using the IN operator.

Query

Query History

Scratch Pad X

1

SELECT *

2

FROM customers

3

WHERE country IN ('Germany', 'France', 'UK')

Data Output

Messages

Notifications

Q4. Find all orders placed in the year 1997.

Query

Query History

```

1 SELECT *
2 FROM orders
3 WHERE order_date BETWEEN '1997-01-01' AND '1997-12-31'

```

Data Output

Messages

Notifications

+

📄

▼

📄

▼

🗑️

🔍

⬇️

📶

SQL

Showing rows

	order_id [PK] smallint	customer_id character	employee_id smallint	order_date date	required_date date	shipped_date date	ship_via smallint	freight real	ship_name character varying (40)
1	10400	EASTC	1	1997-01-01	1997-01-29	1997-01-16	3	83.93	Eastern Connection
2	10401	RATTC	1	1997-01-01	1997-01-29	1997-01-10	1	12.51	Rattlesnake Canyon Grocery
3	10402	ERNSH	8	1997-01-02	1997-02-13	1997-01-10	2	67.88	Ernst Handel
4	10403	ERNSH	4	1997-01-03	1997-01-31	1997-01-09	3	73.79	Ernst Handel
5	10404	MAGAA	2	1997-01-03	1997-01-31	1997-01-08	1	155.97	Magazzini Alimentari Riuniti
6	10405	LINOD	1	1997-01-06	1997-02-03	1997-01-22	1	34.82	LINO-Delicatesses
7	10406	QUEEN	7	1997-01-07	1997-02-18	1997-01-13	1	108.04	Queen Cozinha
8	10407	OTTIK	2	1997-01-07	1997-02-04	1997-01-30	2	91.48	Ottilies Käseladen
9	10408	FOLIG	8	1997-01-08	1997-02-05	1997-01-14	1	11.26	Folies gourmandes
10	10409	OCEAN	3	1997-01-09	1997-02-06	1997-01-14	1	29.83	Océano Atlântico Ltda.
11	10410	BOTTM	3	1997-01-10	1997-02-07	1997-01-15	3	2.4	Bottom-Dollar Markets
12	10411	BOTTM	9	1997-01-10	1997-02-07	1997-01-21	3	23.65	Bottom-Dollar Markets
13	10412	WARTH	8	1997-01-13	1997-02-10	1997-01-15	2	3.77	Wartian Herkku
14	10413	LAMAI	3	1997-01-14	1997-02-11	1997-01-16	2	95.66	La maison d'Asie
15	10414	FAMIA	2	1997-01-14	1997-02-11	1997-01-17	3	21.48	Familia Arquibaldo
16	10415	HUNGC	3	1997-01-15	1997-02-12	1997-01-24	1	0.2	Hungry Coyote Import Store
17	10416	WARTH	8	1997-01-16	1997-02-13	1997-01-27	3	22.72	Wartian Herkku
18	10417	SIMOB	4	1997-01-16	1997-02-13	1997-01-28	3	70.29	Simons bistro
19	10418	QUICK	4	1997-01-17	1997-02-14	1997-01-24	1	17.55	QUICK-Stop
20	10419	RICSU	4	1997-01-20	1997-02-17	1997-01-30	2	137.35	Richter Supermarkt
21	10420	WELLI	3	1997-01-21	1997-02-18	1997-01-27	1	44.12	Wellington Importadora
22	10421	QUEDE	8	1997-01-21	1997-03-04	1997-01-27	1	99.23	Que Delícia
23	10422	FRANS	2	1997-01-22	1997-02-19	1997-01-31	1	3.02	Franchi S.p.A.
24	10423	GOURL	6	1997-01-23	1997-02-06	1997-02-24	3	24.5	Gourmet Lanchonetes
25	10424	MEFEP	7	1997-01-23	1997-02-20	1997-01-27	2	370.61	Mère Paillard

Total rows: 408

Query complete 00:00:00.188

Q5. Display all products where the discontinued status is false.

Query

Query History

Scratch Pad

```

1  SELECT *
2  FROM products
3  WHERE discontinued=0

```

Data Output

Messages

Notifications

+

📄

📄

📄

🗑️

🔍

📶

SQL

Showing rows: 1 to 67

Page 1

	product_id [PK] smallint	product_name character varying (40)	supplier_id smallint	category_id smallint	quantity_per_unit character varying (20)	unit_price real	units_in_stock smallint	units_on_order smallint	reorder_level smallint	discontinued integer
1	3	Aniseed Syrup	1	2	12 - 550 ml bottles	10	13	70	25	0
2	4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22	53	0	0	0
3	6	Grandma's Boysenberry Spread	3	2	12 - 8 oz jars	25	120	0	25	0
4	7	Uncle Bob's Organic Dried Pears	3	7	12 - 1 lb pkgs.	30	15	0	10	0
5	8	Northwoods Cranberry Sauce	3	2	12 - 12 oz jars	40	6	0	0	0
6	10	Ikura	4	8	12 - 200 ml jars	31	31	0	0	0
7	11	Queso Cabrales	5	4	1 kg pkg.	21	22	30	30	0
8	12	Queso Manchego La Pastora	5	4	10 - 500 g pkgs.	38	86	0	0	0
9	13	Konbu	6	8	2 kg box	6	24	0	5	0
10	14	Tofu	6	7	40 - 100 g pkgs.	23.25	35	0	0	0
11	15	Genen Shouyu	6	2	24 - 250 ml bottles	13	39	0	5	0
12	16	Pavlova	7	3	32 - 500 g boxes	17.45	29	0	10	0
13	18	Carnarvon Tigers	7	8	16 kg pkg.	62.5	42	0	0	0
14	19	Teatime Chocolate Biscuits	8	3	10 boxes x 12 pieces	9.2	25	0	5	0
15	20	Sir Rodney's Marmalade	8	3	30 gift boxes	81	40	0	0	0
16	21	Sir Rodney's Scones	8	3	24 pkgs. x 4 pieces	10	3	40	5	0
17	22	Gustaf's Knäckebröd	9	5	24 - 500 g pkgs.	21	104	0	25	0
18	23	Tunnbröd	9	5	12 - 250 g pkgs.	9	61	0	25	0
19	25	NuNuCa Nuß-Nougat-Creme	11	3	20 - 450 g glasses	14	76	0	30	0
20	26	Gumbär Gummibärchen	11	3	100 - 250 g bags	31.23	15	0	0	0
21	27	Schoggi Schokolade	11	3	100 - 100 g pieces	43.9	49	0	30	0
22	30	Nord-Ost Matjeshering	13	8	10 - 200 g glasses	25.89	10	0	15	0
23	31	Gorgonzola Telino	14	4	12 - 100 g pkgs	12.5	0	70	20	0
24	32	Mascarpone Fabioli	14	4	24 - 200 g pkgs.	32	9	40	25	0
25	33	Geitost	15	4	500 g	2.5	112	0	20	0

Total rows: 67

Query complete 00:00:00.120

Part 2: Arithmetic and String Operators

Q1. Show a list of products with a new column that displays unit price increased by 15%.

Query

Query History

1

SELECT product_name,

2

unit_price,

3

unit_price*1.15 AS increased_price

4

FROM products

Data Output

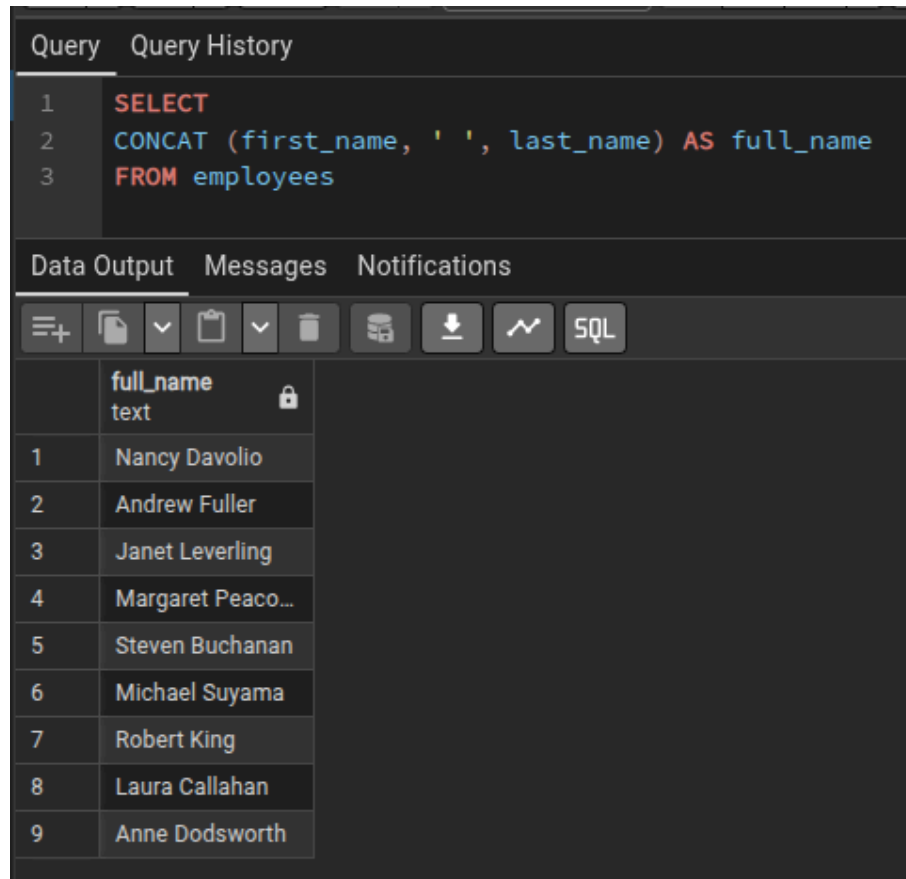
Messages

Notifications

SQL

	product_name character varying (40)	unit_price real	increased_price double precision
1	Chai	18	20.7
2	Chang	19	21.849999999999998
3	Aniseed Syrup	10	11.5
4	Chef Anton's Cajun Seasoning	22	25.299999999999997
5	Chef Anton's Gumbo Mix	21.35	24.552500438690185
6	Grandma's Boysenberry Spread	25	28.749999999999996
7	Uncle Bob's Organic Dried Pears	30	34.5
8	Northwoods Cranberry Sauce	40	46
9	Mishi Kobe Niku	97	111.55
10	Ikura	31	35.65
11	Queso Cabrales	21	24.15
12	Queso Manchego La Pastora	38	43.699999999999996
13	Konbu	6	6.8999999999999995
14	Tofu	23.25	26.737499999999997
15	Genen Shouyu	13	14.95
16	Pavlova	17.45	20.06750087738037
17	Alice Mutton	39	44.849999999999994
18	Carnarvon Tigers	62.5	71.875
19	Teatime Chocolate Biscuits	9.2	10.579999780654907
20	Sir Rodney's Marmalade	81	93.14999999999999
21	Sir Rodney's Scones	10	11.5
22	Gustaf's Knäckebröd	21	24.15
23	Tunnbröd	9	10.35
24	Guaraná Fantástica	4.5	5.175
25	NuNuCa Nuß-Nougat-Creme	14	16.099999999999998
Total rows: 77		Query complete 00:00:00.116	

Q2. Retrieve the full name of employees by concatenating FirstName and LastName.



The screenshot shows a SQL IDE interface with a dark theme. The top section has tabs for 'Query' and 'Query History'. The 'Query' tab is active, displaying a SQL query across three lines:

```
1 SELECT
2 CONCAT (first_name, ' ', last_name) AS full_name
3 FROM employees
```

Below the query editor, there are tabs for 'Data Output', 'Messages', and 'Notifications'. The 'Data Output' tab is active, showing a table of results. Above the table is a toolbar with icons for adding, saving, and deleting queries, as well as a 'SQL' button. The table has two columns: 'full_name' (text) and a lock icon. It contains nine rows of employee names.

	full_name text
1	Nancy Davolio
2	Andrew Fuller
3	Janet Leverling
4	Margaret Peaco...
5	Steven Buchanan
6	Michael Suyama
7	Robert King
8	Laura Callahan
9	Anne Dodsworth

Q3. Find all products where the product name contains the letter 'a' at least twice.

Query		Query History	
1	SELECT		
2	product_id,		
3	product_name		
4	FROM products		
5	WHERE product_name ILIKE '%a%a%'		
6			
Data Output		Messages	Notifications
≡+ 📄 ▼ 📄 ▼ 🗑 🗑 📄 📄 📄 SQL			
	product_id [PK] smallint	product_name character varying (40)	
1	4	Chef Anton's Cajun Seasoning	
2	6	Grandma's Boysenberry Spread	
3	7	Uncle Bob's Organic Dried Pears	
4	8	Northwoods Cranberry Sauce	
5	11	Queso Cabrales	
6	12	Queso Manchego La Pastora	
7	16	Pavlova	
8	18	Carnarvon Tigers	
9	19	Teatime Chocolate Biscuits	
10	20	Sir Rodney's Marmalade	
11	24	Guaraná Fantástica	
12	25	NuNuCa Nuß-Nougat-Creme	
13	28	Rössle Sauerkraut	
14	32	Mascarpone Fabioli	
15	34	Sasquatch Ale	
16	37	Gravad lax	
17	40	Boston Crab Meat	
18	41	Jack's New England Clam Cho...	
19	42	Singaporean Hokkien Fried Mee	
20	44	Gula Malacca	
21	47	Zaanse koeken	
22	49	Maxilaku	
23	50	Valkoinen suklaa	
Total rows: 37		Query complete 00:00:00.115	

Q4. Calculate the total revenue (UnitPrice * Quantity) for each order in the OrderDetails table.

Query

Query History

```

1  Select
2  order_id,
3  SUM(unit_price*quantity) AS total_revenue
4  FROM order_details
5  GROUP BY order_id
6

```

Data Output

Messages

Notifications

≡

📄

▼

📋

▼

🗑️

🗄️

⬇️

📈

SQL

	order_id smallint 🔒	total_revenue double precision 🔒
1	11038	750.9999980926514
2	10782	12.5
3	10725	287.7999954223633
4	10423	1020
5	10518	4150.050006866455
6	10356	1106.3999938964844
7	10963	68
8	10596	1476.1000366210938
9	10282	155.40000534057617
10	10658	4667.999973297119
11	10283	1414.8000202178955
12	10579	317.75
13	10693	2333.999988555908
14	10896	750.5
15	10660	1701
16	10253	1444.7999839782715
17	10425	480
18	10774	875
19	10615	120
20	10514	8623.4499168396
21	10850	740
22	10276	419.99999046325684
23	10978	1500.7000007629395
Total rows: 830		Query complete 00:00:00.126

Q5. Find all suppliers whose company name contains the word "food" (case insensitive)

Query

Query History

Scratch Pad X

1

2

3

4

SELECT *

FROM suppliers

WHERE company_name ILIKE '%food%'

Data Output

Messages

Notifications

Showing rows: 1 to 1

Page No: 1 of 1

	supplier_id [PK] smallint	company_name character varying (40)	contact_name character varying (30)	contact_title character varying (30)	address character varying (60)	city character varying (15)	region character varying (15)	postal_code character varying (10)	country character varying (15)
1	19	New England Seafood Cann...	Robb Merchant	Wholesale Account Ag...	Order Processing Dept. 2100 Paul Revere Bl...	Boston	MA	02134	USA

Part 3: Special Operators (IN, BETWEEN, LIKE, IS NULL, etc.)

Q1. Retrieve all orders where the Freight cost is between 50 and 100.

[illegible]

Q2. Find all orders where the ShipRegion is NULL.

Query

Query History

Scratch Pad

1

SELECT *

2

FROM orders

3

WHERE ship_region IS NULL

Data Output

Messages

Notifications

≡

📄

▼

📄

▼

🗑️

📄

📄

📄

📄

SQL

Showing rows: 1 to 507

Page No: 1 of 1

	order_id [PK] smallint	customer_id character	employee_id smallint	order_date date	required_date date	shipped_date date	ship_via smallint	freight real	ship_name character varying (40)	ship_address character varying (60)	ship_city character varying (15)	ship_region character varying (15)
1	10248	VINET	5	1996-07-04	1996-08-01	1996-07-16	3	32.38	Vins et alcools Chevalier	59 rue de l'Abbaye	Reims	[null]
2	10249	TOMSP	6	1996-07-05	1996-08-16	1996-07-10	1	11.61	Toms Spezialitäten	Luisenstr. 48	Münster	[null]
3	10251	VICTE	3	1996-07-08	1996-08-05	1996-07-15	1	41.34	Victuailles en stock	2, rue du Commerce	Lyon	[null]
4	10252	SUPRD	4	1996-07-09	1996-08-06	1996-07-11	2	51.3	Suprêmes délices	Boulevard Tirou, 255	Charleroi	[null]
5	10254	CHOPS	5	1996-07-11	1996-08-08	1996-07-23	2	22.98	Chop-suey Chinese	Hauptstr. 31	Bern	[null]
6	10255	RICSU	9	1996-07-12	1996-08-09	1996-07-15	3	148.33	Richter Supermarkt	Starenweg 5	Genève	[null]
7	10258	ERNSH	1	1996-07-17	1996-08-14	1996-07-23	1	140.51	Ernst Handel	Kirchgasse 6	Graz	[null]
8	10259	CENTC	4	1996-07-18	1996-08-15	1996-07-25	3	3.25	Centro comercial Moctezuma	Sierras de Granada 9993	México D.F.	[null]
9	10260	OTTIK	4	1996-07-19	1996-08-16	1996-07-29	1	55.09	Ottilies Käseladen	Mehrheimerstr. 369	Köln	[null]
10	10263	ERNSH	9	1996-07-23	1996-08-20	1996-07-31	3	146.06	Ernst Handel	Kirchgasse 6	Graz	[null]
11	10264	FOLKO	6	1996-07-24	1996-08-21	1996-08-23	3	3.67	Folk och få HB	Åkergatan 24	Bräcke	[null]
12	10265	BLONP	2	1996-07-25	1996-08-22	1996-08-12	1	55.28	Blondel père et fils	24, place Kléber	Strasbourg	[null]
13	10266	WARTH	3	1996-07-26	1996-09-06	1996-07-31	3	25.73	Wartian Herkku	Torikatu 38	Oulu	[null]
14	10267	FRANK	4	1996-07-29	1996-08-26	1996-08-06	1	208.58	Frankenversand	Berliner Platz 43	München	[null]
15	10270	WARTH	1	1996-08-01	1996-08-29	1996-08-02	1	136.54	Wartian Herkku	Torikatu 38	Oulu	[null]
16	10273	QUICK	3	1996-08-05	1996-09-02	1996-08-12	3	76.07	QUICK-Stop	Taucherstraße 10	Cunewalde	[null]
17	10274	VINET	6	1996-08-06	1996-09-03	1996-08-16	1	6.01	Vins et alcools Chevalier	59 rue de l'Abbaye	Reims	[null]
18	10275	MAGAA	1	1996-08-07	1996-09-04	1996-08-09	1	26.93	Magazzini Alimentari Riuniti	Via Ludovico il Moro 22	Bergamo	[null]
19	10276	TORTU	8	1996-08-08	1996-08-22	1996-08-14	3	13.84	Tortuga Restaurante	Avda. Azteca 123	México D.F.	[null]
20	10277	MORGK	2	1996-08-09	1996-09-06	1996-08-13	3	125.77	Morgenstern Gesundkost	Heerstr. 22	Leipzig	[null]
21	10278	BERGS	8	1996-08-12	1996-09-09	1996-08-16	2	92.69	Berglunds snabbköp	Berguvsvägen 8	Luleå	[null]
22	10279	LEHMS	8	1996-08-13	1996-09-10	1996-08-16	2	25.83	Lehmanns Marktstand	Magazinweg 7	Frankfurt a.M.	[null]
23	10280	BERGS	2	1996-08-14	1996-09-11	1996-09-12	1	8.98	Berglunds snabbköp	Berouvsåöen 8	Luleå	[null]

Total rows: 507

Query complete 00:00:00.166

LF

Ln 3, Col 26

Q3. Get a list of customers whose contact name starts with 'A' and ends with 'n'.

Query Query History

```
1 SELECT *
2 FROM customers
3 WHERE contact_name LIKE 'A%n'
```

Data Output Messages Notifications

SQL

	customer_id [PK] character	company_name character varying (40)	contact_name character varying (30)
1	EASTC	Eastern Connection	Ann Devon

Q4. Find all employees hired between 1990-01-01 and 1995-12-31.

Query Query History

```
1 SELECT *
2 FROM employees
3 WHERE hire_date BETWEEN '1990-01-01' AND '1995-12-31'
```

Data Output Messages Notifications

SQL

	employee_id [PK] smallint	last_name character varying (20)	first_name character varying (10)	title character varying (30)	title_of_courtesy character varying (25)	birth_date date	hire_date date
1	1	Davolio	Nancy	Sales Representative	Ms.	1948-12-08	1992-05-01
2	2	Fuller	Andrew	Vice President, Sales	Dr.	1952-02-19	1992-08-14
3	3	Leverling	Janet	Sales Representative	Ms.	1963-08-30	1992-04-01
4	4	Peacock	Margaret	Sales Representative	Mrs.	1937-09-19	1993-05-03
5	5	Buchanan	Steven	Sales Manager	Mr.	1955-03-04	1993-10-17
6	6	Suyama	Michael	Sales Representative	Mr.	1963-07-02	1993-10-17
7	7	King	Robert	Sales Representative	Mr.	1960-05-29	1994-01-02
8	8	Callahan	Laura	Inside Sales Coordina...	Ms.	1958-01-09	1994-03-05
9	9	Dodsworth	Anne	Sales Representative	Ms.	1966-01-27	1994-11-15

Q5. Retrieve all products that are either out of stock (UnitsInStock = 0) or have fewer than 10 units remaining.

Query

Query History

Scratch Pad

1

2

3

4

```
SELECT *
FROM products
WHERE units_in_stock < 10
```

Data Output

Messages

Notifications

Part 4: Bitwise and Pattern Matching Operators

Q1. Retrieve all products where the ProductName starts with 'Ch' and has exactly 7 characters.

The screenshot shows a SQL query editor with the following query:

```
1 SELECT *
2 FROM products
3 WHERE product_name LIKE 'ch_____'
```

Below the query editor, the 'Data Output' tab is active, displaying a table with the following columns and data types:

product_id	product_name	supplier_id	category_id	quantity_per_unit	unit_price	units_in_stock	units_on_order	reorder_level	discontinued
[PK] smallint	character varying (40)	smallint	smallint	character varying (20)	real	smallint	smallint	smallint	integer

Q2. Find all employees whose EmployeeID is an even number using a bitwise operator.

The screenshot shows a SQL query editor with the following query:

```
1 SELECT *
2 FROM employees
3 WHERE (employee_id & 1)=0;
```

Below the query editor, the 'Data Output' tab is active, displaying a table with the following columns and data:

employee_id	last_name	first_name	title	title_of_courtesy	birth_date	hire_date	address
[PK] smallint	character varying (20)	character varying (10)	character varying (30)	character varying (25)	date	date	character varying (60)
2	Fuller	Andrew	Vice President, Sales	Dr.	1952-02-19	1992-08-14	908 W. Capital Way
4	Peacock	Margaret	Sales Representative	Mrs.	1937-09-19	1993-05-03	4110 Old Redmond Rd.
6	Suyama	Michael	Sales Representative	Mr.	1963-07-02	1993-10-17	Coventry House\nMiner ...
8	Callahan	Laura	Inside Sales Coordina...	Ms.	1958-01-09	1994-03-05	4726 - 11th Ave. N.E.

Q3. Get all customers whose phone number starts with "(503)".

Query

Query History

Scratch Pad

1

SELECT *

2

FROM customers

3

WHERE phone LIKE '(503)%'

Data Output

Messages

Notifications

Q4. Fetch all orders where the ShipPostalCode contains only numeric digits.

Query		Query History	
1	SELECT *		
2	FROM orders		
3	WHERE ship_postal_code ~ '^[0-9]+'		

Scratch Pad x

Showing rows: 1 to 669 Page No: 1

Showing rows: 1 to 669   Page No: 1

Showing rows: 1 to 669 Page No: 1

Total rows: 669 Query complete 00:00:00.176